7. Shape Calculator

Program :

```
from abc import ABC, abstractmethod
import math

# Abstraction
class Shape(ABC):

    @abstractmethod
    def area(self):
        pass

# Inheritance
class Rectangle(Shape):

    def __init__(self, length, breadth):
        self.length = length
        self.breadth = breadth

    # Polymorphism (Overriding)
    def area(self):
        return self.length * self.breadth

class Circle(Shape):

    def __init__(self, radius):
        self.radius = radius
```

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```
    # Polymorphism (Overriding)
    def area(self):
        return math.pi * self.radius * self.radius

# ---- Main Program ----

choice = input("Enter Shape (Rectangle/Circle): ")

if choice.lower() == "rectangle":
    length = float(input("Enter Length: "))
    breadth = float(input("Enter Breadth: "))
    shape = Rectangle(length, breadth)
    print("Area of Rectangle:", shape.area())

elif choice.lower() == "circle":
    radius = float(input("Enter Radius: "))
    shape = Circle(radius)
    print("Area of Circle:", round(shape.area(), 2))

else:
    print("Invalid Shape")
```

Output :

```
Enter Shape (Rectangle/Circle): Circle
Enter Radius: 10
Area of Circle: 314.16
```

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